Math 540

Homework 5

Name: Gianluca Crescenzo

Section 26

Exercise 1.

(a) *Let* $\mathcal{T}$ *and* $\mathcal{T}'$ *be two topologies on the set* X*; suppose that* $\mathcal{T}' \supset \mathcal{T}$*. What does compactness of* X *under one of these topologies imply about compactness under the other?*

(b) *Show that if* X *is compact Hausdorff under both* $\mathcal{T}$ *and* $\mathcal{T}'$*, then either* $\mathcal{T}$ *and* $\mathcal{T}'$ *are equal or they are not comparable.*

Proof. (a) If X is compact under $\mathcal{T}'$, then it is compact under $\mathcal{T}$. Indeed, if $\mathcal{U}$ is an open cover of X in $\mathcal{T}$, then $\mathcal{U}$ is also an open cover of X in $\mathcal{T}'$. Since X is compact under $\mathcal{T}'$, then $\mathcal{U}$ contains a finite subcover of X. Since our choice of $\mathcal{U}$ was arbitrary, X must be compact under $\mathcal{T}$. However, if X is compact under $\mathcal{T}$, we do not necessarily have compactness under $\mathcal{T}'$. Consider $\mathbb{R}$ with the discrete topology and the trivial topology. The trivial topology is courser than the discrete topology, and $\mathbb{R}$ is compact under the trivial topology, however $\mathbb{R}$ is *not* compact under the discrete topology. Indeed, $\{\{x\} | x \in X\}$ is an open cover of X which admits no finite subcover.

(b) Suppose $\mathcal{T} \subseteq \mathcal{T}'$. Consider the identity map $I : (X, \mathcal{T}') \to (X, \mathcal{T})$. Since it is a map from a compact space into a Hausdorff space, I must be a homeomorphism. Therefore $\mathcal{T} = \mathcal{T}'$. □

Exercise 2(a). *Show that in the finite complement topology on* $\mathbb{R}$*, every subspace is compact.*

Proof. Let $A \subseteq \mathbb{R}$. Suppose $\{U_i\}_{i \in I}$ is an open cover of A. Note that the complement of each open cover will be a finite set. Consider the set $A \setminus U_j$. If $A \setminus U_j = \emptyset$, then U_j completely contains A, hence it is a finite subcover of $\{U_i\}_{i \in I}$. If $A \setminus U_j \neq \emptyset$, then $A \setminus U_j$ must be finite, as $A \setminus U_j = A \cap U_j^c$. Let $\{x_1, ..., x_n\} := A \setminus U_j$. Since each $x_1, ..., x_n \in A \subseteq \bigcup_{i \in I} U_i$, we can find $U_1, ..., U_n \in \{U_i\}_{i \in I}$ such that $x_j \in U_j$ for each $j = 1, ..., n$. Then by construction $A \cap \bigcap_{j=1}^n U_j^c = \emptyset$, hence $A \subseteq \bigcup_{j=1}^n U_j$. Thus A is compact. □

Exercise 5. *Let* A *and* B *be disjoint compact subspaces of the Hausdorff space* X*. Show that there exist disjoint open sets* U *and* V *containing* A *and* B*, respectively.*

Proof. For each $b \in B$, there exists opens sets U_b, V_b such that $b \in U_b$, $A \subseteq V_b$, and $U_b \cap V_b = \emptyset$. Consider the cover $B \subseteq \bigcup_{b \in B} U_b$. Since B is compact, there exists a finite subcover such that $B \subseteq \bigcup_{i=1}^n U_{b_i}$. Since $A \subseteq V_{b_i}$ for each b_i, then $A \subseteq \bigcap_{i=1}^n V_{b_i}$. All together, we've found open sets $\bigcup_{i=1}^n U_{b_i}, \bigcap_{i=1}^n V_{b_i}$ such that $A \subseteq \bigcap_{i=1}^n V_{b_i}$, $B \subseteq \bigcup_{i=1}^n U_{b_i}$, and $\left(\bigcap_{i=1}^n V_{b_i}\right) \cap \left(\bigcup_{i=1}^n U_{b_i}\right) = \emptyset$. □

Exercise 6. *Show that if* $f : X \to Y$ *is continuous, where* X *is compact and* Y *is Hausdorff, then* f *is a closed map.*

Proof. Let C be a closed set in X. Then C is compact. Since f is continuous, $f(C)$ is compact. Since Y is Hausdorff, $f(C)$ is closed. Thus f is a closed map. $\square$

Exercise 7. *Show that if* Y *is compact, then the projection* $\pi_1 : X \times Y \to X$ *is a closed map.*

Proof. Let C be a closed subset of $X \times Y$. Let $x \in X \setminus \pi_1(C)$. Then the slice $\{x\} \times Y$ is disjoint from C. Because Y is compact, by the Tube Lemma, there is a neighborhood W of x such that the whole tube $W \times Y$ is disjoint from C. Therefore, W is a neighborhood of x which is disjoint from $\pi_1(C)$. Hence $X \setminus \pi_1(C)$ is open, which means $\pi_1(C)$ is closed. $\square$

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Exercise 2. *Let* X *be a metric space with metric* d*; let* $A \subset X$ *be nonempty.*

(a) Show that $d(x, A) = 0$ *if and only if* $x \in \overline{A}$.

(b) Show that if A *is compact,* $d(x, A) = d(x, a)$ *for some* $a \in A$.

Proof. (a) Let $U \subseteq X$ be an open set. Suppose $d(x, A) = 0$. We can find a sequence $(a_n)_n \subset A$ such that $d(x, a_n) \to 0$. Hence $a_n \to x$. Find $N \in \mathbb{N}$ sufficiently large such that $a_N \in U$. Hence $U \cap A \neq \emptyset$. Since U was arbitrary, $x \in \overline{A}$. Conversely, let $x \in \overline{A}$ and $\epsilon > 0$. We can find a sequence $(a_n)_n \subset A$ such that $a_n \to x$. Find $N \in \mathbb{N}$ sufficiently large such that $d(x, a_N) < \epsilon$. Then $\inf\{d(x, a) \mid a \in A\} \leqslant d(x, a_N) < \epsilon$, hence $d(x, A) < \epsilon$. Since $\epsilon > 0$ was arbitrary, $d(x, A) = 0$.

(b) Let $x \in X$. Consider the map $f : A \to X$ given by $f(a) = d(x, a)$. Then f is continuous, and $f(A) = \{d(x, a) \mid a \in A\}$. By the Extreme Value Theorem, there exists some $a' \in A$ such that $d(x, a') = \inf\{d(x, a) \mid a \in A\} = d(x, A)$. $\square$

Exercise 3. *Recall that* $\mathbb{R}_K$ *denoted* $\mathbb{R}$ *in the* K*-topology.*

(a) Show that $[0, 1]$ *is not compact as a subspace of* $\mathbb{R}_K$.

(b) Show that $\mathbb{R}_K$ *is connected. (Hint* $(-\infty, 0)$ *and* $(0, \infty)$ *inherit their usual topologies as subspaces of* $\mathbb{R}_K$*)*

(c) Show that $\mathbb{R}_K$ *is not path connected.*

Proof. (a) Recall that the K-topology is generated by the basis of open intervals (a, b) together with all sets of the form $(a, b) \setminus K$. Define:

$$U_n := \left(\tfrac{1}{n}, 2\right) \cup \Big((-1, 1) \setminus K\Big), \ n \in \mathbb{N}.$$

Then $\{U_n\}_{n \in \mathbb{N}}$ is an open cover which does not admit a finite subcover.

(b) Since $(-\infty, 0)$ and $(0, \infty)$ inherit their usual topologies as subspaces of $\mathbb{R}_K$, they are connected. Hence their closures are connected; i.e., the sets $(-\infty, 0]$ and $[0, \infty)$ are connected. Since their intersections are nonempty, their union is connected as well, which equals $\mathbb{R}_K$.

$\square$

Exercise 4. *Show that a connected metric space having more than one point is uncountable.*

Proof. Let (X, d) be a connected metric space. Let $x_0, x_1 \in X$. Define $f : X \to \mathbb{R}$ by:

$$f(x) = \frac{d(x_0, x)}{d(x_0, x) + d(x_1, x)}.$$

Clearly f is continuous, and $f(x_0) = 0$ and $f(x_1) = 1$. Since the continuous image of a connected space is connected, it must be the case that $f(X) = [0, 1]$. Since $[0, 1]$ is uncountable, X is uncountable. $\square$